5 VISUALISATION, DESIGN, AND CREATION OF COMPONENTS

Learning Outcomes

The students will be able to:

- understand the mechanical block of a robot and its applications.
- visualise the robotic components using Tinkercad.
- design different robotic joints with the help of Tinkercad.
- visualise the motion of different joints using Tinkercad.

The mechanical block of a robot creates its physical structure. It includes components that make robots work in a real-world scenario. The functions of robots are reflected with the help of this block, or this block helps robots interact with their environment.

Applications of Mechanical Block

Some important applications of the mechanical block of a robot are:

Logistics: A robot with a proper mechanical block can be used for logistics and warehousing operations. The mechanical block of these kinds of robots should include mechanisms to pick, place, and move the wheels. It performs tasks like order arranging, picking, sorting, and packing items.

Agriculture: These robots should have mechanisms for planting seeds, controlling weeds, and harvesting crops to perform different agricultural tasks.

Medical robotics: The mechanical part of these robots should use tools that are very precise and accurate in medical diagnosis, surgeries, etc.

Manufacturing robots: The mechanical block of these robots should be capable of performing different manufacturing tasks, like handling material, product integration, welding, quality checking, etc.

Household work robots: Robots designed to perform household work should have mechanical tools to perform tasks, like picking and placing objects, cleaning, moving, etc.

Components of the Mechanical Block of a Robot

The important components of the mechanical block of a robot are:

Outer body: It is also called the chassis and creates the physical framework of a robot. It allows integration of different mechanical components of the robot. It gives look to a robot. The design of the body varies on the basis of applications, like it may be a wheeled robot, a humanoid, a cleaning robot, etc.

Joints and links: Joints help robots to move. Joints and links together create the structure of a robot.

Actuators: Different actuators, like motors and hydraulic devices, are used to drive joints.

End effectors: These are the tools used to interact with the outer environment. Based on t application, an end effector may be a tool for cutting, gripping, painting, mopping, etc.

Controller: The mechanical block is controlled by a controller. The task of the controller is accept input from the sensor and send signals to other components and blocks of the robot.

Activity 5.1

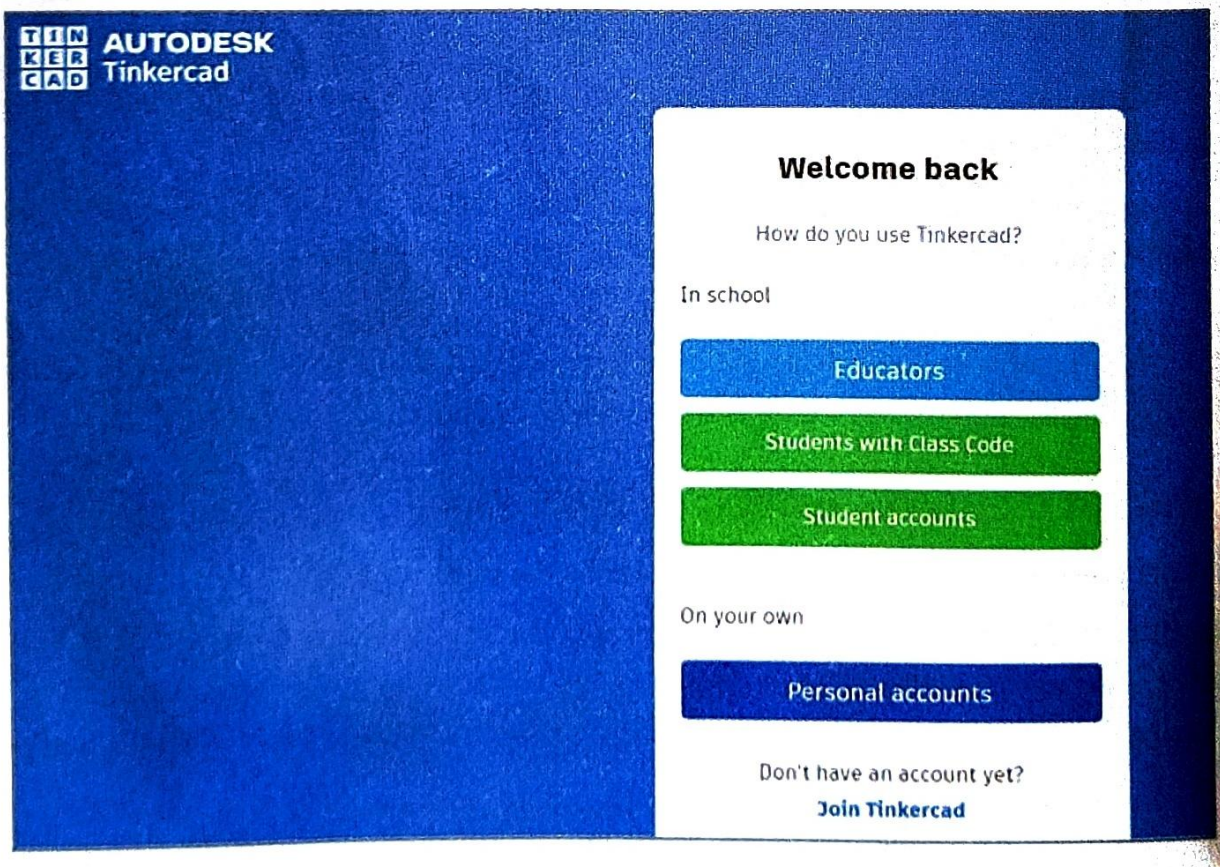
Identify the different part of this robotic arm as components of the mechanical body (joints, links, end effectors, etc.).



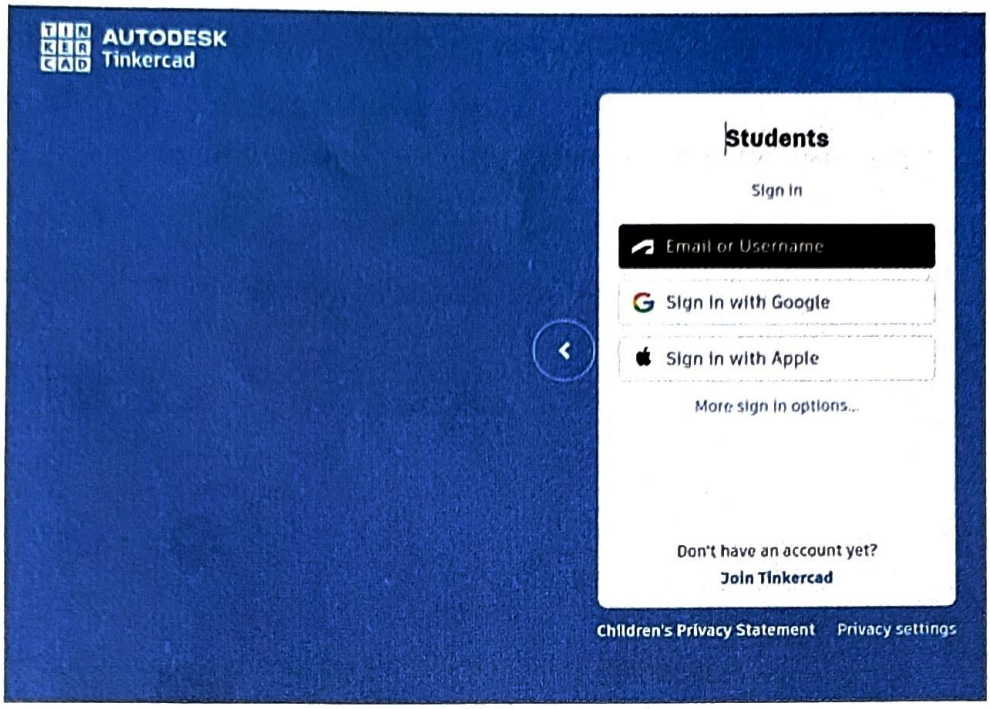
Visualising the Components of a Robot

Tinkercad is a freely available online software program developed by **Autodesk.** It is used visualise, design, and create circuits. It provides 3D design of different components, whi helps in better visualisation and understanding. It also provides support for the basic robo design. To use Tinkercad for visualisation, follow these steps:

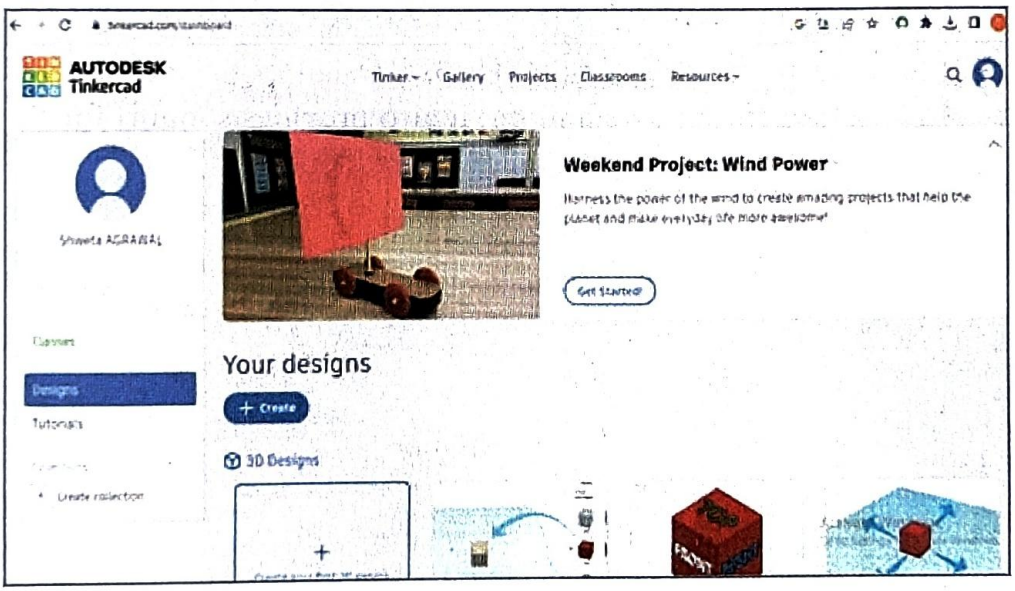
Step 1: Login to Tinkercad (https://www.tinkercad.com/login).



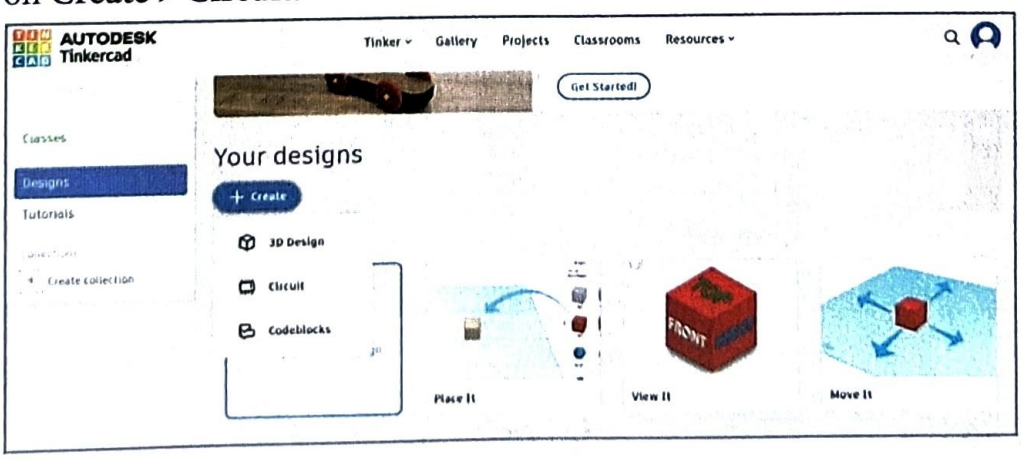
Step 2: Select the **Student accounts** option and login with Google, or create account through **Join Tinkercad**.



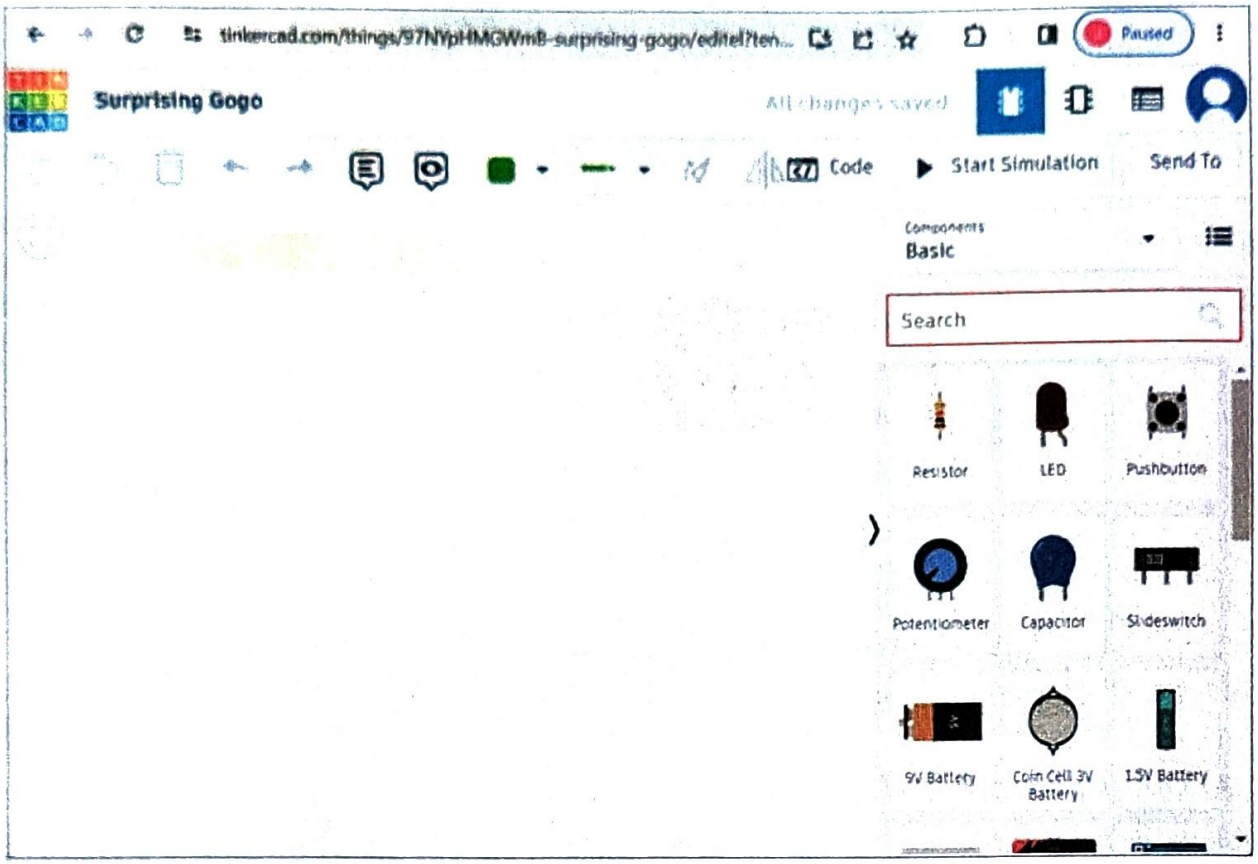
Step 3: After login, click on the **Designs** option from the left pane. You will get the following screen.



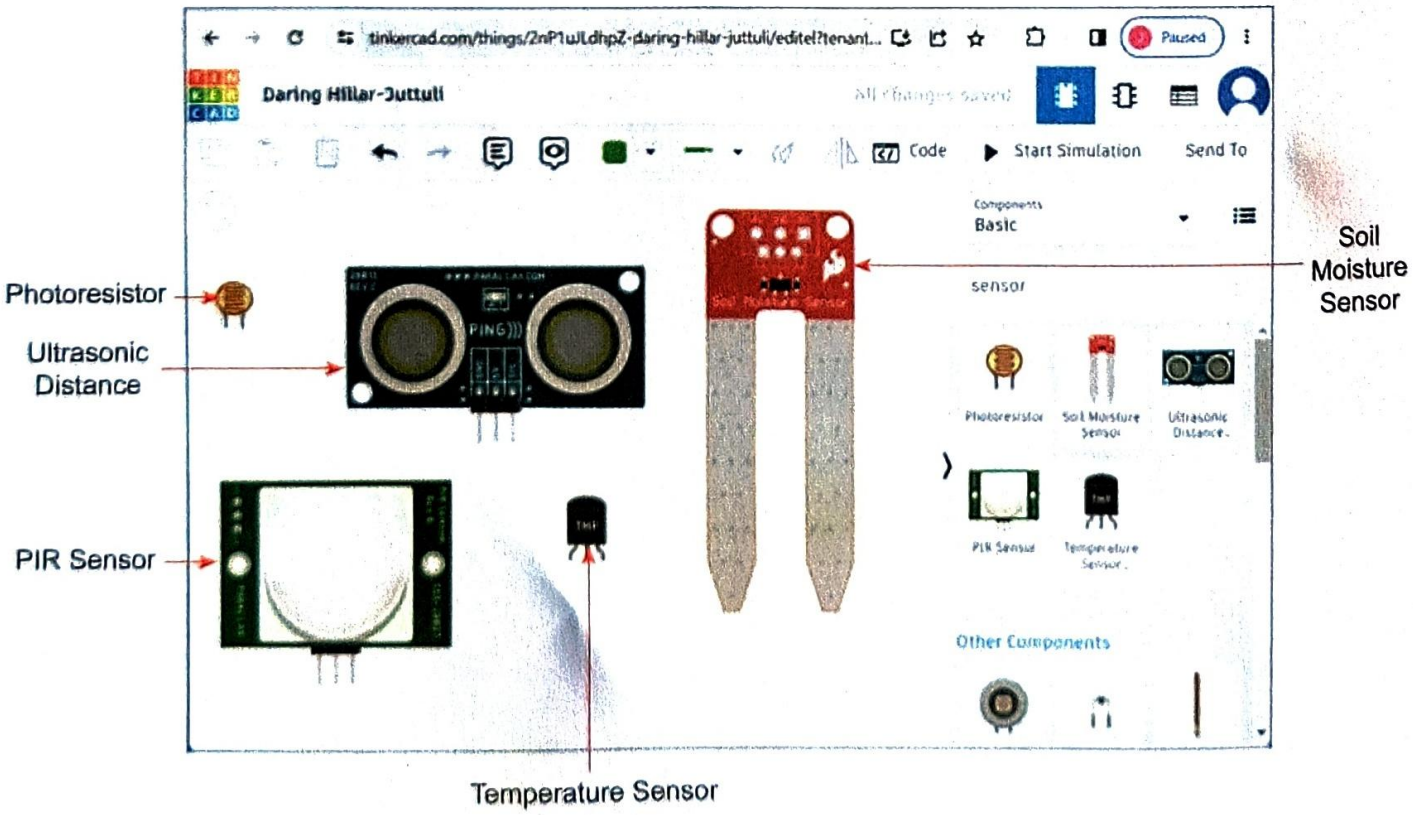
Step 4: Click on **Create > Circuit**.



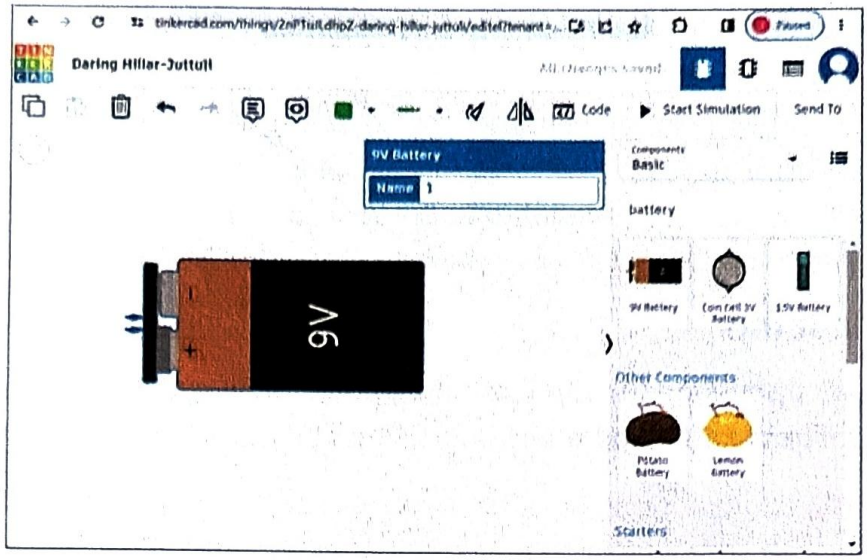
Step 5: In the next screen, click on the **Search** option, given at right side, and type the component that you want to use or visualise.



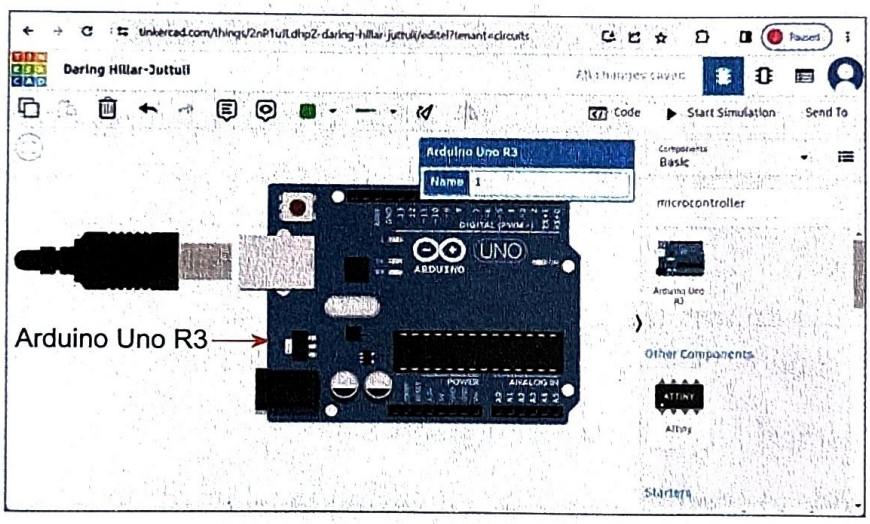
Step 6: Type '**sensor**' in the search field, which will show different types of sensors. The scree given below shows five sensors: **Photoresistor** (light sensor), **Soil Moisture Sensor, Ultrasoni Distance, PIR Sensor,** and **Temperature Sensor**.



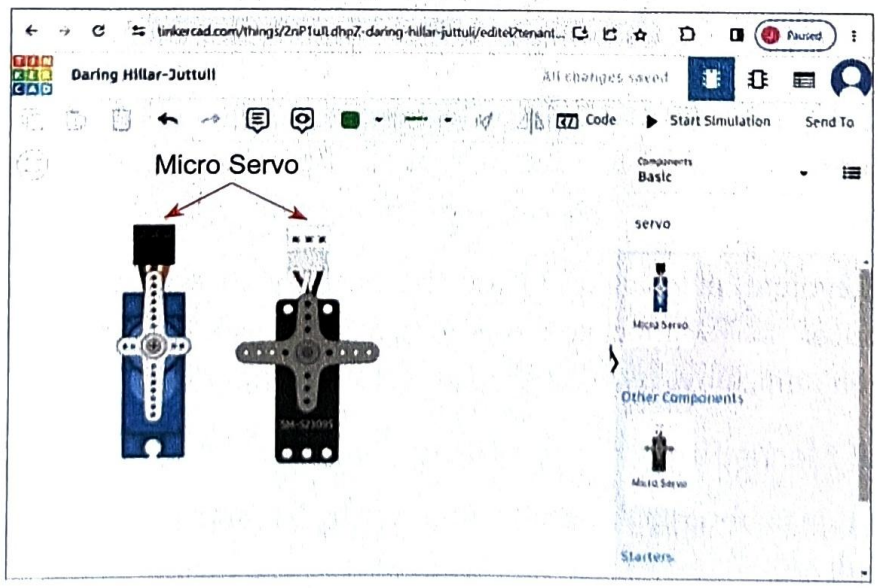
Step 7: Now, search for '**battery**', which will show different types of batteries. The commonly used battery is **9V Battery**, which is shown below.



Step 8: Search for '**microcontroller**', which will show **Arduino UnoR3** (the most common microcontroller for designing robots).



Step 9: Search for '**servo**', which will show **Micro Servo** motors.



Activity 5.2

Visualise the different batteries available in Tinkercad.

Activity 5.3

Visualise the different motors available in Tinkercad.

Types of Robotic Joints

In this chapter, we are discussing the three robotic joints, namely, Revolute Joint, Prismatic Joint, and RR Mechanism. The creation and movement of these joints are explained below.

Revolute Joint

Revolute joints, also known as rotary joints, allow the links to rotate around an axis. These joints are commonly found in robotic arms and legs, and they allow the robot to move its joints in different directions.

Visualisation of Revolute Joint Using Tinkercad

Scan the given QR code to observe the design for creating a revolute joint with the help of Tinkercad.

Prismatic Joint

Prismatic joints are also known as linear joints. These joints allow the links to move in a straight line along an axis. A common example of a prismatic joint is the sliding mechanism in a drawer.

Visualisation of Prismatic Joint Using Tinkercad

Scan the given QR code to observe the design for creating a prismatic joint with the help of Tinkercad.

Activity 5.4

Create a prismatic joint with the help of Tinkercad.

RR Mechanism

The RR (Revolute-Revolute) mechanism means the use of two revolute joints in series. The most common application of the RR mechanism is in a robotic arm. The two revolute joints can be placed at the shoulder and elbow, respectively, in a robotic arm.

Visualisation of RR Mechanism Creation Using Tinkercad

Scan the given QR code to observe the design for creating RR mechanism with the help of Tinkercad.

ReCap

- The mechanical block of a robot provides its outer body and helps in interacting with its outer environment.
- Some of the applications of the mechanical block of a robot are logistics, safety, movement, etc.
- Tinkercad is an open source platform for 3D design and basic robotics understanding.
- Different components of robots, like sensors, batteries, controllers, etc., can be visualised and used in Tinkercad.

Brain Developer

Section A: Solved Questions

A. State True or False.

1. The task of a mechanical block is to provide the body to a robot.
2. The controller controls the functioning of a robot.
3. Joints and links are the mechanical parts of the body of a robot.
4. Prismatic joint represents the motion on multiple axis.
5. Tinkercad allows the 3D designing of the mechanical parts of the body of a robot.
6. Tinkercad is developed by Amazon.
7. Tinkercad can be used as a desktop application.
8. The RR mechanism consists of one revolute and one prismatic joint.

B. Select the correct option.

1. Arduino Uno is an example of ..... .
   a. Sensor
   b. Actuator
   c. Controller
   d. Motor

2.  Image is representing ..... .
   a. IR sensor
   b. Ultrasonic sensor
   c. Solar sensor
   d. Temperature sensor

3.  Image is representing ..... .
   a. Micro servo
   b. Hydraulic motor
   c. Water motor
   d. None of these

4. Tinkercad allows..... .
   a. Only 2D designs
   b. Only architectural designs
   c. 3D designs and simulations
   d. Only text-based designs

5. Which of the following is an important advantage of Tinkercad?

a. It requires advanced programming skills
b. A user-friendly platform
c. Very complex simulation features
d. None of these

6. Which of the following is responsible for the mechanical part of a robot?

a. Giving shape to the robot
b. Allowing interaction with outer environment
c. Protecting robot's electronic part
d. All of the above

7. Revolute joint shows rotation along ..... .

a. Single axis
b. Multiple axis
c. Either a or b
d. None of these

8.  is representing a ..... .

a. Humidity sensor
b. Temperature sensor
c. Transistor
d. None of these

C. Fill in the blanks.

1. The ..... are the tools used to interact with the outer environment.
2. ..... is very important in medical robots.
3. The end effector for a painter robot would be a ..... .
4. Tinkercad can be used for 3D designing and ..... .

Answer Key

A.	1. True	2. False	3. True	4. False	5. True
	6. False	7. False	8. False		
B.	1. c	2. b	3. a	4. c	5. b
	6. d	7. a	8. b		
C.	1. End effectors	2. Precision	3. Paint brush	4. Simulation	

D. Short answer type questions.

1. What do you understand by revolute joint?

Ans. Revolute joints, also known as rotary joints, allow the links to rotate around an axis. These joints are commonly found in robotic arms and legs, and they allow the robot to move its joints in different directions.

2. What do you understand by prismatic joint?

Ans. Prismatic joints are also known as linear joints. These joints allow the links to move in a straight line along an axis. A common example of a prismatic joint is the sliding mechanism in a drawer.

3. What do you understand by RR mechanism?

Ans. The RR (Revolute-Revolute) mechanism means the use of two revolute joints in series. The most common application of the RR mechanism is in a robotic arm. The two revolute joints can be placed at the shoulder and elbow, respectively, in a robotic arm.

E. Long answer type questions.

1. What are the various components of the mechanical block of a robot?

Ans. The important components of the mechanical block are:

Outer body: It is also called as chassis and creates the physical framework of a robot. It allows integration of different mechanical components of a robot. It provides look to the robot. The design of body varies on the basis of applications, like it may be a wheeled robot, humanoid, cleaning robot, etc.

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Actuators: Different actuators, like motor and hydraulic devices are used to drive joints.

End effectors: These are the tools used to interact with outer environment. Based on the application, an end effector may be a tool for cutting, gripping, painting, mopping, etc.

Controller: This block is controlled by a controller. The task of controller is to accept input from sensor and send signals to other components and blocks of the robot.

2. What are the various applications of mechanical block of a robot?

Ans. Some important applications are:

Logistics: A robot with a proper mechanical block can be used for logistics and warehousing operations. The mechanical block of these kinds of robots should include mechanisms to pick and place the object and move the wheels. It performs the tasks like order arranging, picking, sorting, and packing items.

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Manufacturing robots: The mechanical block of these robots should be capable to perform different tasks, like handling material, product integration, welding, quality checking, etc.

Household work robots: Robots designed to perform household work should have mechanical tools to performs tasks like picking and placing an object, cleaning, moving, etc.

Section B: Unsolved Questions

F. Short answer type questions.

1. What is Tinkercad?
2. Write the names of the sensors available in Tinkercad.
3. Write the names of the batteries available in Tinkercad.

G. Create 3D design for the following in Tinkercad.

1. Revolute Joint
2. Prismatic Joint
3. RR Mechanism

Higher Order Thinking Skills (HOTS)

Critical Thinking

1. Create a robotic arm with threes Degrees of Freedom by using Tinkercad.
2. Explore Tinkercad to design a robotic arm for the task of planting a tree.